

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 397 – SENIOR DESIGN II
Spring 2026**

Lecture:

M/W: 1 pm-2:15 pm

No Classes:

March 23rd and 25th (Spring break)

Office Hours:

Mondays and Wednesdays 12.15 pm-1.00 pm (Dr. Bilgin)

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teach Assistant:

None

Text:

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler)

Course Description:

This course builds on the skills taught in CHE 396. Design methodology and principles will be applied to the design of large-scale chemical processes and plants. The design needs to be analyzed from process simulation, safety, environmental impact, controls, optimization, feasibility, and economics. Competing processes must be critically examined and presented. An essential part of this course is working in teams effectively. The selection of design topics is structured to be flexible to open up opportunities for creativity. Since the course is structured to encourage flexible approaches to design problems, intermediate presentations and reports are required to keep students on track and evaluate their design feasibility.

Course Objectives:

This course involves the application of all prior CHE concepts to the chemical process design. The design needs to be analyzed from safety, environmental impact, feasibility, risk, and economics perspectives. The course aims to equip students with the skill to evaluate open-ended design problems conceptually, analytically, and practically.

Learning Objectives :

At the end of the course, students should be able to:

- Work effectively with team members
- Present effectively technical material based on the audience
- Identify key components in market analysis and construct design basis
- Analyze competing processes
- Write technical content in a way that is appropriate to the audience
- Develop a process flow diagram
- Simulate full process with Aspen Plus or SuperPro
- Size process equipment
- Identify control strategies for the plant
- Carry out economic analysis
- Identify potential safety concerns in the process and formulate strategies to mitigate those

Requirements/Constraints:

- All the calculations need to be submitted in excel
- All the work needs to be archived on Blackboard. Work needs to be shared with group mentors via Microsoft Teams, Google Drive, or Box account.
- There will be four reports throughout the semester.
- There will be four presentations throughout the semester. During presentations, all the group members need to participate.
- The final report needs to be submitted to the mentors two weeks before to the end of the semester.
- Seven steppingstones for the project
 1. Process description
 2. Material/energy balances and PFDs
 3. More detailed equipment calculations (at this point, student are encouraged to contact vendors)
 4. Pricing equipment, economics, cash flows
 5. Control scheme
 6. Safety/potential hazardous materials list
 7. Final report

Report/Presentation Deliverables:

Reports should have the following sections: Introduction, Main body (process description, process control, process safety, and environmental analysis), Summary (executive summary and recommendations), and Appendix (design basis, BFD, PFD, material balance, sample calculations, equipment list and sizing, utilities, economic calculations). Listed below are the deliverables for reports. Presentation deliverables are the same as Report deliverables in a PowerPoint presentation format.

Report 1:

- *Introduction:* This section informs everyone about the project definition and its significance. It should address why this project is important and why the reader should get interested to learn more about the project.
 - Problem definition and significance

- Market analysis (supply-demand analysis, global share, value chain and pricing of raw materials, and products)
- Judging criteria (more environmentally friendly, more economical, etc.)
- *Process description*: This section gives detailed information on alternative ways to process and a detailed explanation of the chosen process.
 - Competing processes (each competing process needs to be explained in detail, including Block Flow Diagram, advantages and disadvantages of the process)
 - Chosen process (explain each unit in detail, including operating conditions, critically analyze alternatives for each unit, example: possible catalysts, advantages and disadvantages of each, list your assumptions)
- *Appendix*
 1. *Design Basis*: Clear explanation of how you choose the plant's size, the amount of the raw materials needed, and location of the plant.
 2. *Block Flow Diagram (BFD)*:
 3. *Preliminary Process Flow Diagram (PFD)*: Only major equipment needs to be included.
 4. *Rough Material Balance*: Mass balance for block in- block out, estimated amounts of products, by-products, waste, etc.

Report 2:

The process needs to be simulated with Aspen or SuperPro. Following deliverables need to be added to Report 1 material in the Appendix section and submitted as Report 2.

- *Final PFD*: All the equipment needs to be shown and streams need to be numbered.
- *Detailed mass balance*: Detailed inputs and outputs to each part of the process along with conditions ie, temperature, pressure, etc.
- *Equipment list*: Most of the major equipment should be identified along with a good start on the equipment list.
- *Equipment sizing*: Column sizes, pump and compressor sizing, heat exchanger sizing, along with horsepower requirements, and heating/cooling loads.
- *Sample Hand Calculation*: One distillation tower sizing or separation unit sizing should be done by hand calculation. Hand calculation and Aspen results should be compared and analyzed critically.
- *Preliminary utilities*: A summary of the utilities demand of the entire plant.

Report 3:

The process needs to be analyzed in detail with the focus on controls, safety, environment, and economics. Following deliverables need to be added to Report 2 material in the main section and submitted as Report 3. All the detailed calculations need to be added to the Appendix section.

- *Process controls analysis*: Describe your control objectives and control strategy for the entire process units. Draw a control loop schematic for one unit in the process and explain it in detail.
- *Process safety and environmental analysis*: Analyze chemical and operational safety. List environmental issues, air emissions, water discharges, etc. Describe design features to mitigate the effects of these discharges. Outline the process hazards which you must legally and ethically address. (That means, for instance, all OSHA PSM highly hazardous materials should be listed, including quantities on site. Process hazards associated with

toxic, explosive, hot and cold process streams should be identified and a safeguard included in the design.). Carry out a HAZOP analysis for one unit of your process. It should be the same unit as the one you draw process control loops.

- *Process economics analysis*: Calculate capital cost, operational cost, simple payback time, ROI. Analyze cost in detail (NPV, DCFROR, sensitivity analysis, etc.). Discuss the meaning of your results rather than just listing the numbers.

Report 4 (Final Report):

Previous reports need to be corrected/ updated based on the mentors' and the professor's feedback. The following sections need to be added to the report and submitted.

- *Executive summary*: A 1-2 pages summary for the company's high-level executives, such as the CEO or company president, who might not necessarily have a high level of technical process knowledge. This section should include a brief explanation of the market analysis, amount of product, location of the plant, cost analysis, and recommendations.
- *Recommendations*: Overall conclusion of the process. Describe your concerns with the process and state if you recommend or not to move forward. Explain the uncertainties and risks involved and future steps.
- *Utilities demand*: Finalize the utility demand for the entire plant.
- *Heat exchanger network(optional)*: This part is optional. If your focus is on energy consumption, you should perform pinch analysis and include heat exchanger network design.

Assigning groups

Groups are assigned at the end of the Fall semester, and each group needs to submit a proposal by the end of the Fall semester. Groups are assigned as students choosing one person who they want to work with and one who they do not want to work with. Based on their choices, Dr. Bilgin assigns the group with four or five members in each group. Each group has an industry mentor assigned by Dr. Bilgin. All the groups must contact their mentor at the end of the Fall semester and get the mentor's approval on their project topic before submitting their proposal.

Team Policies and Responsibilities

This course is designed to complete a project with your team. Your team will have a number of responsibilities as it completes deliverables listed above:

- *Prepare a group contract*. This should include the leader's list and responsibilities, internal due dates, communication method and expectations, consequences and policies to fire someone from the group. Students who are fired or who quit will get zeroes for the remaining assignments. Submit your group contract on Blackboard under your Group Blog tab.
- *Designate roles within the team*. This could include the leader, editor, librarian (who keeps up with the resources). Each group is responsible for choosing a group leader and rotating the leader among the group members every three weeks. The leader's responsibility is to set up the meetings with the mentor and manage the group.
- *Participate in mentor meetings*. Each group is responsible for meeting with their mentor weekly (physical meeting or online meeting, or phone conversation).

- *Submit meeting agenda and meeting minutes.* Before the meeting, each group must submit a meeting agenda to the corresponding mentor two days before the meeting. If the agenda is not submitted, then the meeting will be canceled. Each group must submit meeting notes to the corresponding mentor the next day after the meeting. Both meeting agenda and meeting minutes need to be submitted on Blackboard under the Group Blog tab.
- *Mark your calendar for presentation days.* Each team will have 15 minutes to present and 10 minutes for questions. There are six groups in the class. Group presentations will be 1 pm-2.15 pm on Monday/ Wednesday of the week marked as presentation week in the course outline below as announced on the course Blackboard site. Everyone must attend entire presentations. Students missing presentation day will be marked absent and that will count as two unexcused absences.
- *Meet and review your report and presentation slides before submission.* Everyone should equally contribute to the project and be familiar with the details of the project. Also, everyone needs to submit their work to the group by the team's internal due date.
- *Review feedback on the report and presentations.* Everyone in the group should review the feedback given by the professor, mentors, and classmates.
- *Complete and submit online peer evaluations.* After each deliverable, each individual will be asked to complete a peer evaluation. This will help to identify conflicts within the team early in the semester.

Written reports and presentations are the product of team efforts, and it is expected that each member contributes equally. Normally each team member will receive the same grade for each deliverable. However, if the peer evaluation percentage is less than %80, individual grades will be calculated by multiplying team grades with peer evaluation percentages.

Course Outline:

Date	Concept	Assignment
January 12 th 2026	Syllabus review	
January 14 th 2026	Group work	
January 19 th 2026	NO CLASS (MLK day)	
January 21 st 2026		
January 26 th 2026		
January 28 th 2026		
February 2 nd 2026		
February 4 th 2026		
February 9 th 2026		
February 11 th 2026		Report 2 is due Saturday midnight
February 16 th 2026	Second presentation	and Ppt 2 is due Sunday midnight
February 18 th 2026	Second presentation	
February 23 rd 2026	NO CLASS	
February 25 th 2026		
March 2 nd 2026		

March 4 th 2026		
March 9 th 2026		
March 11 th 2026		Report 3 is due Saturday midnight
March 16 th 2026	Third presentation	and Ppt 3 is due Sunday midnight
March 18 th 2026	Third presentation	
March 23 rd 2026	NO CLASS	
March 25 th 2026	NO CLASS	
March 30 th 2026		
April 1 st 2026		
April 6 th 2026	Final presentation	Final presentation is due Sunday midnight
April 8 th 2026	Final presentation	
April 13 th 2026		
April 15 th 2026		
April 20 th 2026		
April 22 nd 2026	NO CLASS	
April 24 th 2026	EXPO	
April 27 th 2026	Senior Design Showcase	
April 29 th 2026		
May 6 th 2026		Final Report is due

Exams:

There will not be any exam.

Attendance:

Students are expected to attend all classes, and attendance is incorporated as 5% of the final grade. Students are allowed two excused absences OR one unexcused absence. If absences exceed these limits, the student will receive a zero for attendance. Absences on the days of presentations count as two unexcused absence.

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance:	5%
Presentation 2:	10%
Report 2:	10%
Presentation 3:	10%
Report 3:	10%
Final Presentation:	12.5%
Final Report:	12.5%
CHE EXPO poster:	5%
CHE EXPO presentation:	10%
Meeting with Mentors:	10%
SACHE modules:	5%

SaChE safety modules:

Students are expected to complete two SaChE safety modules online and submit their certificates of completion to the instructor. Students can choose two of the modules listed under Level Three Curriculum on this website: <https://www.aiche.org/ccps/education/safety-and-chemical-engineering-education-sache-certificate-program>.

Academic Integrity Policy

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <http://dos.uic.edu/community-standards/>

Academic Misconduct

Please see collaboration policies written above for each type of work for this class. In all cases, you cannot submit someone else's work (partial or complete) as your own. You are guilty of academic misconduct if you extend or receive any kind of unauthorized assistance. Absolutely no transfer of assignment between students is permitted (paper or electronic) under any circumstance, and you may not solicit code from other people, family, friends, or internet resources. Other examples of academic misconduct include emailing your work to another student, copying-pasting from the internet, working in a group on an assignment that states it is individual work, and allowing a tutor, TA, or another individual to write an answer or code for you. Academic misconduct is unacceptable. Minimum penalty will be a zero score on that assignment or assessment, with a letter drop to your final grade in the class. I reserve the right to exercise a change in the way grades are calculated if I deem it is necessary. More than one academic misconduct offense in this course will result in failure of the class. This means if you took this course a previous semester and were found guilty of academic misconduct, a second offense this semester will result in failure from the course. Similarly, if you have two offenses in the same semester, you will receive an F in course. Note that the Dean of Students will change your grade from W to an F and only with my approval can it remain as a W (if you were found guilty and drop the class). EVERY academic misconduct offense must be reported to the Dean of Students (if you ask me not to, you are asking me to not do my job, so please don't).

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure: <http://policies.uic.edu/educational-policy/student-academic-grievance-policy/>

Disability Accommodation

The University of Illinois at Chicago is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities at UIC. Students with disabilities who require accommodations for access to and/or participation in this course are welcome, but must be registered with the Disability Resource Center (DRC). You may contact DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay) and consult the following: <http://drc.uic.edu/guide-to-accommodations>. If you have a letter of accommodation, it is best to stop by my office hours to talk about it in person. Any exams scheduled in the DRC must overlap with the originally scheduled exam. In other words, you must start the exam before or at the same time as the originally scheduled exam. Additionally, you may not leave the DRC exam room until after the originally scheduled exam is over or later.

Mental Health and Emotional Wellness

We value your mental health and emotional wellness as part of the UIC student experience. The UIC Counseling Center offers an array of services to provide additional support throughout your time at UIC, including workshops, peer support groups, counseling, self-help tools, and initial consultations to speak to a mental health counselor about your concerns. Please visit the Counseling Center website for more information (<https://counseling.uic.edu/>). Further, if you think emotional concerns may be impacting your academic success, please contact your faculty and academic advisers to create a plan to stay on track.

Campus Advocacy Network

If you have experienced sexual assault, domestic/dating violence, family abuse, stalking, or harassment, there is help available on campus. The Campus Advocacy Network provides free and confidential services to UIC students, faculty, and staff who have experienced or are experiencing interpersonal violence. If you would like to connect with an advocate, please contact CAN at can-appointment@uic.edu or call (312) 413-8206. To learn more, visit CAN's website at <https://can.uic.edu/>.